



SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)

(Sponsored by Sri Vasavi Educational Society)
(Approved by AICTE, New Delhi & Recognized by UGC under section 2(f) & 12(B))
(Permanently affiliated to JNTUK, Kakinada, Accredited by NBA and NAAC with 'A' Grade)
Pedatadepalli, TADEPALLIGUDEM-534 101.W.G.Dist. (A.P)

Department of Computer Science and Engineering

A.Y. 2026-27

Project Title: Machine Learning Based Solar Energy Prediction and Monitoring System

Abstract:

Solar output changes constantly with the weather and the sun's position, making it hard to know how much energy a panel will produce at a given place and time. Most tools show weather data without turning it into an energy figure, or expect technical knowledge ordinary users lack.

This project presents a machine learning system for solar prediction and monitoring, delivered through a simple web interface. The user picks a location by city name or GPS coordinates and sets a date and time. The system finds the matching latitude and longitude and collects solar irradiance, air temperature and wind speed. These go to four models — Random Forest, Histogram Gradient Boosting, Extremely Randomised Trees and Ridge Regression — combined by a stacking ensemble that estimates output in kilowatt-hours. Training uses years of hourly weather history for that location, with matching energy output computed from a standard photovoltaic performance model.

Results appear with an interactive map and automatic insights, such as whether conditions are safe for operation and how wind is cooling the panels. Tree-based ensembles handle the non-linear link between weather and output, which fixed formulas capture poorly; Ridge Regression is kept as a linear baseline so the added complexity must prove its worth. Each model was tested on unseen data held out in date order using R^2 and mean absolute error, and accuracy is reported with the conditions where it drops, mainly fast-changing cloud cover. It gives users an accessible way to estimate output, monitor performance and decide on installations.

Keywords: Solar prediction; Machine learning; Random Forest; Histogram Gradient Boosting; Extremely Randomised Trees; Ridge Regression; Stacking ensemble; Solar irradiance; Weather data; Photovoltaic performance model

COs Addressed: CO1, CO2, CO3, CO4, CO5

Pos & PSOs Addressed: PO1, PO2, PO3, PO4, PO5, PO6, PSO1, PSO2

Quality of the Project:

- **Factors:** Environment, Sustainability, Cost, Ethics
- **Type:** Application, Product & Research.
- **Quality:** A location-aware solar prediction platform that couples a four-model stacking ensemble with a standard photovoltaic performance model, validated on chronologically held-out data and delivered through a web interface with interactive mapping and automated operational insights.



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Mapped SDGs:

- SDG - 07: Affordable and Clean Energy
- SDG - 09: Industry, Innovation and Infrastructure
- SDG - 13: Climate Action

Submitted By: CSE-D_13

S. No	Roll No.	Name of the Student
1	23A81A050P5	T. Harshini Sowmya
2	23A81A05M9	M. Lavanya Durga
3	23A81A05O3	P. J.S Siddhartha
4	23A81A05M8	M. Ajay Prakash
5	23A81A05O4	P. Maruthi SreeRam
6	23A81A05L9	K. Surya Prakash

Supervisor

Head of the Department

List of COs for Main Project

- V23CSP03.1** Assess a real-world problem and formulate its requirements specifications. (K5)
- V23CSP03.2** Analyze tools, algorithms, and techniques that contribute to the software solution of the project. (K4)
- V23CSP03.3** Evaluate structural or architectural designs for a given set of system requirements. (K5)
- V23CSP03.4** Utilize collaborative teamwork principles to function effectively as a responsible member or leader in developing software solutions. (K3)
- V23CSP03.5** Organize technical documentation in written reports and deliver effective oral presentations. (K3)

List of POs for UG Engineering Programs:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems / components / processes to meet identified needs with



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consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming.

PSO2: Use Professional Engineering practices and strategies for development and maintenance of software.

Factors

Environment, Sustainability, Safety, Ethics, Cost, etc.

Type



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Application, Product, Research, Review, etc.

Quality of Product

(Write a comment to your work)

SDG 'S are:

1. No poverty
2. Zero hunger
3. Good health and well - being
4. Quality education
5. Gender Equality
6. Clean water and sanitation
7. Affordable and clean energy
8. Decent work and economic growth.
9. Industry, innovation, and infrastructure
10. Sustainable cities and communities
11. Reduced inequality
12. Responsible consumption and production
13. Climate action
14. life below the river
15. Life on land
16. Peace, justice, and strong institution
17. Partnership for the goals.